Lisa Blum Moyse

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Experience

Postdoctoral position in the Integrative Biophysics Group: Foraging models
Supervisor: Ahmed El Hady
15/12/2023 - 15/07/2025,
01/02/2026 - 31/01/2027
University of Konstanz, Centre for the Advanced Study of Collective Behaviour
Konstanz, Germany

Visiting research fellow in the Mathematical, Computational and Experimental Neuroscience group
Collaborators:
Serafim Rodrigues, Mathieu Desroches
01/10/2023 - 30/11/2023
Basque Center for Applied Mathematics
Bilbao, Spain

PhD: Computational neuroscience models at different levels of abstraction for synaptic plasticity, astrocyte modulation of synchronization and systems memory consolidation
Supervisor: Hugues Berry
01/09/2020 - 31/08/2023
Defense: 14/09/2023
INSA Lyon, LIRIS
Lyon, France

Internship: A stochastic model for heterogeneous reactions: the case of enzymatic mineralisation
Supervisors:
Pietro de Anna, Franco Bagnoli
01/06/2019 - 31/08/2019
University of Florence
Florence, Italy

Internship: Imaging in biological environments by wavefront shaping method
Supervisors:
Baptiste Blochet, Sylvain Gigan
01/06/2018 - 31/07/2018
Laboratoire Kastler Brossel
Paris, France

Internship: Misfolded proteins in neurodegenerative diseases
Supervisors:
Benjamin Dehay, Erwan Bezard
01/05/2017 - 30/06/2017
Institut des Maladies Neurodégénératives
Bordeaux, France

Education

École Normale Supérieure de Lyon, Lyon, France
Master of Physics, 2018 - 2020
Bachelor's degree of Physics, 2017 - 2018
Bachelor's degree of Biology, 2016 - 2017

Lycée Michel Montaigne, Bordeaux, France
Higher school preparatory classes, two-year undergraduate course in Biology, Chemistry, Physics, Mathematics and Geology (BCPST), 2014 - 2016

Publications

Blum Moyse, L., & El Hady, A. (2026). Socio-cognitive models in a patch foraging setting: a case study for model selection and parameter identifiability methods. *arXiv*.
http://arxiv.org/abs/2609.36231

Blum Moyse, L., & El Hady, A. (2026). Social hierarchy shapes foraging decisions. *Physical Review Research*.
https://doi.org/10.1103/5vfb-jg8f

Blum Moyse, L., & El Hady, A. (2025). Social patch foraging theory in an egalitarian group. *PRX Life, 3*(3), 033007.
https://journals.aps.org/prxlife/abstract/10.1103/mqxr-1nyv

Marienhagen J., Blum Moyse L., Schakowski A., Kahl B., Davidson J., El Hady A., Kurvers R. & Deffner D. (2025). Bridging reinforcement-learning and drift-diffusion modeling to uncover the cognitive processes underlying collective foraging. *PsyArXiv*.
https://osf.io/preprints/psyarxiv/a3ptk_v1

Blum Moyse, L., & Berry, H. (2024). A coupled neural field model for the standard consolidation theory. *Journal of Theoretical Biology, 588*, 111818.
https://doi.org/10.1016/j.jtbi.2024.111818

Blum Moyse, L. (2023). Computational neuroscience models at different levels of abstraction for synaptic plasticity, astrocyte modulation of synchronization and systems memory consolidation (Doctoral dissertation, INSA Lyon).
https://theses.hal.science/tel-04390158

Blum Moyse, L., & Berry, H. (2022). Modelling the modulation of cortical Up-Down state switching by astrocytes. *PLoS Computational Biology, 18*(7), e1010296.
https://doi.org/10.1371/journal.pcbi.1010296

Blochet, B., Joaquina, K., Blum, L., Bourdieu, L., & Gigan, S. (2019). Enhanced stability of the focus obtained by wavefront optimization in dynamical scattering media. *Optica, 6*(12), 1554-1561.
https://doi.org/10.1364/OPTICA.6.001554

Communications

Mechanistic models of social foraging (talk)
Journées de la Biologie Théorique 2026
Grenoble, France, 09/11/2026-10/11/2026
https://sfbt.math.cnrs.fr/jbiot2026.html

Mechanistic models of social foraging (talk)
Journées Math-Bio Santé 2026
Lille, France, 02/11/2026-06/11/2026
https://jmbs2026.sciencesconf.org/?lang=fr

Modeling the emergence of collective narratives on Twitter/X from individual decision-making processes (poster)
Social Simulation Conference
Durham, United Kingdom, 24/08/2026 - 28/08/2026
https://www.conftool.pro/ssc2026/sessions.php

Mechanistic theories of social foraging (talk)
Workshop: Circuits, Dynamics, and Computation in Social Behavior.
COSYNE: Computational and Systems Neuroscience, Conference
Cascais, Portugal, 17/03/2026
https://cosyne-social-behavior.github.io/

Social hierachy shapes foraging decisions (talk)
Symposium: Neural circuits and decision strategies for behavioral trade-offs
16th Göttingen Meeting of the German Neuroscience Society
Göttingen, Germany, 28/03/2025
https://www.nwg-goettingen.de/2025/default.asp?id=8

Collective foraging in a two patches environment: theory and applications (poster)
Foraging and Information Seeking Conference
Lyon, France, 11/12/2024 - 13/12/2024
https://sites.google.com/view/foragingconference/home

Collective decisions in social patch foraging (talk)
Satellite workshop of the Bernstein Conference: Algorithms and neural mechanisms of social and collective behavior
Frankfurt-am-Main, Germany, 29/09/2024
https://bernstein-network.de/bernstein-conference/past-future-bernstein-conferences/bernstein-conference-2024/

Modelling Systems Memory Consolidation with neural fields (poster)
COSYNE: Computational and Systems Neuroscience, Conference
Lisbon, Portugal, 03/2022
https://www.cosyne.org/past-conferences

Systems consolidation model: How does the hippocampus guide the reorganization of the information stored in the neocortex (poster)
Statistical Mechanics and Emergent Phenomena in Biology, summer school
Beg Rohu, France, 06/2021
https://www.phys.ens.psl.eu/fr/cours/beg-rohu-summer-school-physics

Mentoring and Teaching

Teaching assistant at the NeuroBridges summer school of computational neuroscience: Supervision of a journal club and students projects, Cluny, *16/09/2024 - 26/09/2024*
`https://neurobridges.net/previous/2024`

Teaching assistant at the Konstanz summer school of collective behaviour: Supervision of students' projects and tutorial, CASCB Konstanz, *22/07/2024 - 09/08/2024*
`https://www.exc.uni-konstanz.de/kscb/previous-summer-schools/2024/`

Supervisor of a bibliographic project in computational neuroscience, ENS de Lyon, *2021*

Physics tutorials and practical work (64h), INSA Lyon, *2020 - 2021*

Collective responsibilities

External reviewer for the conference Complex Networks 2026